Blackwish Protocol Nutrition Diet and Portion Methodology

Research reference | 23 October 2026 UTC | Version 1

Food energy, essential nutrients and portion size are related but different questions. Calories describe energy; nutrient composition describes what else a food supplies; a portion record describes the amount actually selected or eaten. None alone establishes an appropriate diet for an individual.

This reference explains calories, protein and the nine essential amino acids, essential fatty acids, dietary patterns, fiber, selected micronutrients and water. It preserves a reproducible way to calculate food and recipe portions, with exact food forms, source identities and uncertainty.

How to use this reference

Understand the numbers: pages 2-7 separate dietary reference values, population guidance and food-label units. Numerical values are scoped to the stated age and life stage.

Estimate a portion: pages 8-12 preserve food identity, weigh edible amounts, account for recipe yield, and show worked calculations plus a reusable record.

Challenge the reasoning: pages 13-15 contain three distinct Zen desk reviews and a bounded Death Knell counterexample audit. Sources follow on pages 16-18.

Status and boundaries

RESEARCH / INTERNALLY DESK REVIEWED / NOT RATIFIED. Review means source, logic, arithmetic and document checks, not clinical validation, regulatory approval or proof of a health outcome. This is an additive reference; earlier cooking research is preserved.

No personal calorie target, weight-loss plan, disease-specific diet, supplement regimen or safety certificate is supplied. No experiment on people, actual food preparation or laboratory nutrient analysis was performed. Nothing here deploys a Blackwish revision or integrates it into a live system. Reference ratification is separate from implementation and deployment authorization; none is granted here.

Measurement is optional and purpose-bound. It should help understanding, purchasing or consistent recipe division, not turn eating into a moral score or a requirement to track every meal.

Calories and reference values

A food **Calorie is a kilocalorie or kcal**. Calories describe energy, not nutritional completeness. Energy needs vary with age, body size, activity and life stage. The National Academies' 2023 energy equations estimate requirements; they do not measure an individual's exact need. No personal estimate is calculated here. [N5]

2,000 kcal is a US labeling reference, not everyone's requirement. A label serving is also a reporting amount, not a recommended portion. A package may contain multiple servings. Match the labeled basis to the amount actually consumed before interpreting calories or percentages. [P1]

Reference	What it means
RDA	Usual daily intake sufficient for nearly all healthy people in a specified age/sex group; not everyone's exact requirement.
AI	Adequate Intake used when evidence is insufficient for an RDA.
EAR	Estimated requirement of half a specified healthy group; not a universal personal target.
UL	Upper intake level unlikely to cause adverse effects; not an intake goal. No UL does not mean unlimited safety.
DV and percent DV	Standardized label references; they may differ from the applicable RDA or AI.
AMDR	Range of energy shares for a macronutrient, not a minimum gram requirement.

The RDA, AI, EAR and UL belong to the Dietary Reference Intakes framework for generally healthy populations. Disease-specific adjustments and diagnoses require a different assessment. [N1]

Energy factors and macronutrient shares

General factors are approximately **4 kcal/g protein, 4 kcal/g carbohydrate and 9 kcal/g fat**. They are useful arithmetic, not a guarantee of reproducing a label. Specific factors, different treatment of nondigestible carbohydrates or sugar alcohols, and rounding may explain differences. Use the matching source's energy value rather than declaring every mismatch an error. [P3]

Established adult AMDRs are **45-65% carbohydrate, 20-35% fat and 10-35% protein** of energy. They are broad ranges; all percentages still share the same total, and the upper endpoints cannot all be chosen together. They are not an individualized diet prescription. [N2]

Equal-energy foods may provide very different nutrients. A correct calorie total cannot by itself establish adequate essential fats, amino acids, fiber, vitamins or minerals.

Protein and the nine essential amino acids

Protein supplies amino acids used in body proteins and other compounds. The nine indispensable amino acids must be obtained through the diet. "Nonessential" amino acids remain necessary; the term means the body can ordinarily make them. Some become conditionally essential in particular circumstances. [N7]

Nine indispensable amino acids	
Histidine	Isoleucine
Leucine	Lysine
Methionine	Phenylalanine
Threonine	Tryptophan
Valine	All nine matter; this is not a supplement list.

Keep two different protein references distinct

The established National Academies **adult RDA is 0.8 g/kg/day**. For an illustrative 70-kg generally healthy adult, multiplication gives 56 g/day. This is a requirement-based adequacy reference, not a universal optimum, athletic plan or disease-specific instruction. [N2]

Separately, the current US **Dietary Guidelines for Americans 2025-2030** gives a **1.2-1.6 g/kg/day** protein goal, with adjustment for individual calorie requirements. This is the current policy's goal, not evidence that a new National Academies RDA has replaced 0.8 g/kg/day. Do not merge the values into one supposedly definitive requirement. [N3]

Grams and amino acid quality answer different questions

Protein quality considers the indispensable-amino-acid pattern and digestibility. FAO's DIAAS method compares digestible individual amino acids with an age-appropriate reference pattern; the lowest ratio identifies the limiting amino acid. An ingredient score is not a complete health ranking or proof that a mixed diet is adequate. [N8]

Different foods can complement each other's amino-acid patterns. For healthy adults, complementary plant proteins need not be paired at every meal; sufficient protein and variety across the day matter. Animal foods and plant foods including legumes, soy, nuts and seeds can contribute. Amounts still count: a small serving labeled "complete protein" does not necessarily supply a day's needs. [N7,N3]

This reference does not assign isolated amino-acid doses or assume that powders improve an already adequate diet. Greater grams alone do not resolve inadequate energy, limited food variety, allergies or a clinical problem.

Essential fats without a mandatory oil

The essential dietary fatty acids are **linoleic acid or LA, an omega-6 fatty acid**, and **alpha-linolenic acid or ALA, an omega-3 fatty acid**. "Essential" describes a nutrient requirement, not a compulsory cooking oil. Aromatic products called essential oils are a different category, not the nutrients meant here. [N2,N6]

Daily AI	Men age 19-50	Women age 19-50
Linoleic acid	17 g	12 g
Alpha linolenic acid	1.6 g	1.1 g

These source-table categories apply to generally healthy adults who are not pregnant or lactating; other life-stage rows can differ. They are dietary reference amounts, not oil quantities or supplement doses. [N2]

Food sources and omega 3 distinctions

Nuts, seeds and oils such as soybean, corn and safflower can supply LA. Flaxseed, chia, walnuts, canola and soybean oil can supply ALA. No particular oil brand or single food is uniquely required. A food's total fat does not reveal how much LA or ALA it contains. [N2,N6]

EPA and DHA are different omega-3 fatty acids. Fish and certain algae-derived products supply them directly. ALA can be converted to EPA and then DHA, but conversion is limited and variable. Do not use one assured conversion factor for a meal or treat equal grams as interchangeable. NIH reports an ALA AI but no separate National Academies RDA/AI for EPA or DHA; that is not a claim that they lack biological roles. [N6]

Fat type and replacement matter

WHO favors unsaturated fats over saturated and trans fats, with saturated fat replaced by polyunsaturated fat, plant-source monounsaturated fat or carbohydrate from naturally fiber-rich foods. Merely adding oil is not the same intervention as replacing a fat source. Coconut and palm oils also show why "plant oil" does not always mean predominantly unsaturated. [N4]

WHO's ceilings are **no more than 10% of energy from saturated fat and 1% from trans fat**, including industrial and ruminant trans fat. The current US guideline also retains a general saturated-fat ceiling of 10% of energy. These are limits, not required intakes. [N4,N3]

An illustrative 2,000-kcal calculation gives about 22 g saturated fat at 10% using 9 kcal/g. This demonstrates the arithmetic only; it does not establish anyone's calorie need. Oil capsules and high-dose supplements are not a default conclusion.

Dietary patterns carbohydrate and fiber

A dietary pattern is the combination of foods eaten over time. A useful assessment asks whether the pattern supplies adequate energy, protein, essential fats and micronutrients, while fitting the person's food access, preferences, culture and tolerability. No single food can certify the whole pattern.

Carbohydrate quality is more than total grams

Whole grains, pulses, vegetables and fruit provide different fiber and micronutrient packages from sugar-sweetened drinks. WHO's 2023 carbohydrate guideline emphasizes these food groups and recommends **at least 25 g/day of naturally occurring dietary fiber for adults**. [N9]

The established US/Canadian total-fiber AIs for ages **19-50 are 38 g/day for men and 25 g/day for women**, outside pregnancy and lactation. These age/sex-specific references and WHO's adult recommendation have different framing; do not treat them as identical definitions or extend the rows to every life stage. [N2]

Sugar terms are not interchangeable

Total sugars includes naturally present and added sugars. **Added sugars** is a food-label category. WHO's **free sugars** includes added sugars plus those in honey, syrups, fruit juice and juice concentrates; intact fruit is treated differently. WHO recommends free sugars below 10% of energy and notes possible additional benefit at 5% or less. "Natural" honey is not exempt from the definition. [P1,N4]

The current US guideline gives a sodium limit below 2,300 mg/day for ages 14 and older. Sodium mass is not salt mass, and age-specific or clinical adjustments are separate questions. [N3]

Build a pattern rather than a perfect meal

A practical synthesis is to vary vegetables and fruit, whole grains or other fiber-rich staples, legumes and other protein foods, suitable nuts or seeds, and calcium/B12-containing or appropriately fortified foods. Select fats for composition, not because a single oil is mandatory. This is an educational framework, not a proposed menu.

Vegetarian and vegan patterns should be assessed for what they actually supply, not judged by their name. Dependable B12 is a particular planning issue when animal foods are absent; fortification and product labels matter. A mixed diet can also have nutrient gaps. [N10]

Convenience, frozen, canned and fortified foods can serve useful roles; processing or packaging alone is not a complete nutrient profile. Compare the relevant food form, composition and practical fit. The current US policy includes frozen, dried and canned produce with little or no added sugars among options. [N3]

One day or one meal is not the entire pattern. Use repeated information where an assessment of usual intake is needed; do not infer deficiency or a health outcome from one calculated meal. [N1]

Selected micronutrients and their limits

Calories and macronutrients do not replace vitamins and minerals. The selected **RDA** values below are for **ages 19-50**; they are total dietary-intake references, not supplement doses or a complete adequacy checklist. Pregnancy and lactation columns are also for this age range. Source category names are retained. [N10-N14]

Nutrient per day	Men	Women not pregnant or lactating	Pregnancy	Lactation
Vitamin B12 mcg	2.4	2.4	2.6	2.8
Iron mg	8	18	27	9
Calcium mg	1,000	1,000	1,000	1,000
Vitamin D mcg	15	15	15	15
Folate mcg DFE	400	400	600	500

Units: mg = milligrams; mcg = micrograms. DFE = dietary folate equivalents, accounting for differences in availability. A bottle's folic-acid mass is not automatically the same numerical amount as DFE. Vitamin D 15 mcg equals 600 IU; the reference assumes minimal sun exposure. [N13,N14]

Food sources and interpretation

B12: dependable natural sources are animal foods; patterns without them need reliable fortified sources or appropriately selected supplementation. Do not credit unfortified plant foods with dependable B12. [N10]

Iron: plant iron is nonheme and less bioavailable than heme iron; vitamin-C-containing foods improve nonheme absorption. The source gives higher requirements for vegetarian patterns. These rows do not diagnose anemia or authorize taking iron. [N11]

Calcium and vitamin D: dairy, suitably fortified alternatives, some calcium-set tofu and fish with edible bones can contribute calcium. Fortified products differ. Few foods naturally supply much vitamin D; fatty fish and fortified foods matter. Appearance or a "plant milk" name does not establish equivalence. [N12,N13]

Folate and iodine: leafy vegetables, legumes and fortified grains can supply folate. Iodized salt supplies iodine, while specialty salts often are not iodized; check the label rather than increasing salt indiscriminately. [N14,N15]

Other relevant nutrients include potassium, magnesium, zinc and vitamins A, C, E and K. A complete assessment uses the appropriate life-stage references and actual foods, particularly when food groups are omitted. Higher intake is not always better; no dosing or toxicity chart is provided.

Water life stage and safety boundaries

Total water is not an extra plain water quota

US total-water AIs for adults are about **3.7 L/day for men and 2.7 L/day for women**, outside pregnancy and lactation. This includes water from **all beverages and food**, not that quantity of plain water in addition to everything else. Heat, activity, losses and illness can alter needs. A fixed "eight glasses" rule is not a universal physiological requirement. [N16]

Do not force a fluid target from this table. Conditions affecting fluid or electrolyte balance can require clinical guidance; no disease adjustment is supplied here.

Do not scale adult reference values indiscriminately

Infants and children: growth, feeding safety and age-specific nutrient references require their own assessment. Adult macro ranges and gram targets should not simply be scaled down by body size.

Pregnancy and lactation: energy and nutrient needs change in ways not captured by a single calorie multiplier. The micronutrient comparison on page 6 illustrates selected differences but is not a prenatal feeding or supplement plan. [N10-N14]

Older adults and other circumstances: altered appetite, chewing or swallowing difficulty, absorption, medications or chronic disease may require additional assessment. A generic portion record does not resolve these issues. Use the applicable reference table and professional judgment where needed.

Nutrition is not a food safety certificate

The nine major allergens under US food-labeling law are milk, eggs, fish, crustacean shellfish, tree nuts, peanuts, wheat, soybeans and sesame. Other foods can also cause allergy. A nutrient-rich food can be unsuitable for someone allergic to it; substitutions should preserve relevant nutrients without treating this reference as allergy clearance. [N17]

Storage, cooking temperatures, contamination and cross-contact are separate food-safety questions. A calorie estimate does not prove that food is safe to eat. Existing cooking research remains separate and is not overwritten by this nutrition reference.

Supplements require a separate question

FDA does not preapprove dietary supplements for safety and effectiveness before marketing. "Natural" does not guarantee safety, and food-pattern research does not automatically justify oil capsules, amino-acid powders or high-dose vitamins. [N18]

Ask which nutrient, how much in the amount consumed, for which population and outcome, supported by what evidence. A food log alone cannot diagnose deficiency or establish a safe dose. This report provides neither individualized treatment nor supplement safety certification.

A reproducible portion workflow

Keep four fields separate: the **label serving** is a reporting unit, the **portion** is the amount selected or eaten, the **recipe yield** is the finished edible amount, and an **appropriate intake** is a separate individualized question. A label serving is not an instruction to eat that amount. [P1]

Identify before calculating

1. State the purpose. Estimate what was eaten, compare like foods, divide a recipe, or plan food purchasing. Do not silently convert a measurement exercise into a calorie restriction.

2. Identify the exact food. Record product/brand or database ID, retrieval date, preparation, raw versus cooked, dry versus rehydrated, drained versus undrained, and edible part. Include skin, bones, shells, oil, sauce or toppings as applicable. A cooked weight must not inherit a raw-food nutrient record.

3. Retain the nutrient basis. Copy the amount per label serving with its gram weight, or the amount per 100 g. Preserve units and whether the value is declared, analyzed, calculated, estimated or missing. Missing information is not a zero. [P3,P4]

Measure and scale

4. Measure the edible amount. Tare a suitable kitchen scale to remove the container. Record what was actually included. If using cups, spoons or counts, retain a food-specific gram conversion. Equal volumes of different foods need not have equal masses. Hand, plate and visual guides are rough planning tools, not exact calorie measures. [P4,P5]

5. Scale every nutrient independently. For a label: portion amount = stated amount per serving x portion grams / serving grams. For a per-100-g record: portion amount = amount per 100 g x portion grams / 100. Units must match and denominators must be positive. [P4]

6. Record uncertainty. Check scale resolution, food variation, rounding, unmeasured additions and whether the food record matches. Round the reported answer sensibly rather than suggesting laboratory accuracy.

7. Assess adequacy separately. A correctly calculated portion can still be unsuitable for an allergy, medical need or life stage. It also may simply be smaller or larger than someone wants or needs. This method prescribes no universal portion weight.

A rough estimate can be appropriate for a low-stakes question. A documented unknown is more useful than a falsely exact number. Retain the purpose so measurement effort does not exceed what the question requires.

Recipe yield and retained nutrients

A recipe calculation requires two distinct estimates: the nutrients retained in the finished food and the finished edible mass. Cooking changes both in ways that depend on the food, process and nutrient; a universal raw-to-cooked multiplier is not justified. [P6,P7]

Preserve the calculation chain

Ingredient contribution = compatible nutrient amount per 100 g x edible ingredient grams / 100. Record all relevant ingredients, including cooking fat, sauces and toppings.

Retained recipe total = sum of ingredient contributions after appropriate preparation and retention adjustments. Record discarded liquids, fat or food residues and any missing data.

Finished density = estimated retained recipe total / measured finished edible yield in grams.

Portion amount = finished density x actual portion grams. Apply separately to kcal, protein, fat, carbohydrate, fiber and other nutrients of interest.

Choose one compatible accounting route

One route uses matching cooked-food entries with cooked edible weights. Another starts with raw ingredients and applies appropriate food-, process- and nutrient-specific yield and retention factors. These are not interchangeable shortcuts: do not apply a cooking adjustment twice. Published USDA retention factors are historical evidence, not measurements of every new recipe. [P4,P6]

Measure the final yield after the relevant cooking, draining and cooling stage, without the pot. If the dish will lose more water before it is portioned, record the stage or remeasure. As-purchased weight, trimmed edible input and final cooked yield belong in separate fields. USDA Food Buying Guide yields support planning, not a personal adult portion prescription. [P5]

Check the allocation assumption

Proportional division assumes a sufficiently uniform mixture. A stew portion with more meat than another is not necessarily equivalent at the same total weight. Measure components separately or label the homogeneous-mixture approximation.

Water gain or loss changes mass and nutrient concentration without itself adding or removing calories. Discarded oil, starch-containing liquid or food can change total energy; loss of individual vitamins or minerals must not be inferred from water loss alone. [P6,P7]

Keep the previous recipe record when ingredients, preparation or yield change. Create a successor record with the revised inputs and source version so a later reader can reproduce both estimates.

Food composition examples with exact forms

The six records below are USDA FoodData Central **SR Legacy** entries. The underlying final SR release was April 2018; these portal records show 1 April 2019 publication. They were retrieved for this research on 23 October 2026, not newly measured in 2026. Values describe the recorded foods, not every brand or recipe. [F1-F6,SLEG]

Basis: amounts per 100 g edible food. Energy is the record's kcal value, not recalculated from 4/4/9. Macronutrients and fiber are rounded here to one decimal gram. The six names immediately below are the exact record descriptions.

Food and FDC ID	kcal	Protein g	Fat g	Carb g	Fiber g
Raw rice [F1] 168877	365	7.1	0.7	80.0	1.3
Cooked rice [F2] 168878	130	2.7	0.3	28.2	0.4
Cooked black beans [F3] 173735	132	8.9	0.5	23.7	8.7
Hard boiled egg [F4] 173424	155	12.6	10.6	1.1	0.0
Olive oil [F5] 171413	884	0.0	100.0	0.0	0.0
Raw apple with skin [F6] 171688	52	0.3	0.2	13.8	2.4

[F1] Rice, white, long-grain, regular, raw, enriched
[F2] Rice, white, long-grain, regular, enriched, cooked
[F3] Beans, black, mature seeds, cooked, boiled, without salt
[F4] Egg, whole, cooked, hard-boiled
[F5] Oil, olive, salad or cooking
[F6] Apples, raw, with skin (Includes foods for USDA's Food Distribution Program)

What these examples can and cannot show

The raw and cooked rice entries illustrate why food form matters. They are separate database records, not a measured before-and-after cooking experiment and not evidence for a universal rice yield factor.

The oil entry is 100 g fat per 100 g yet lists 884 kcal, not 900. Specific energy factors and calculation conventions can differ from the simple 9 kcal/g approximation. That difference alone does not show a record error. [P3]

These examples are not a complete meal plan, ranking of food quality, list of required foods or prescribed portions. A zero reported in a database is a source value; an absent field remains unknown rather than being converted to zero. [P4]

Worked portion and recipe calculations

A label serving is a multiplier

A fictional label lists 240 kcal and 6 g protein per 60 g. A 90-g portion is 90 / 60 = 1.5 label servings. The estimate is **360 kcal and 9 g protein**. This is synthetic arithmetic, not a recommended amount or a tested food.

Match food form and edible mass

For 150 g of the cooked rice record: 130 x 150 / 100 = **195 kcal**. For 150 g of the separate raw rice record: 365 x 150 / 100 = **547.5 kcal**, approximately 548 kcal. These weights are not interchangeable, and the examples do not describe the same rice before and after cooking. [F1,F2]

For 172 g of the cooked black beans record: 132 x 1.72 = 227.04 kcal, reported as **about 227 kcal**; protein = 8.86 x 1.72 = 15.2392 g, reported as **about 15.2 g**. USDA lists 172 g for one cup of that food. It is a food-specific conversion, not a universal cup weight. [F3]

For 13.5 g of the olive oil record: 884 x 0.135 = **about 119 kcal**. USDA lists 13.5 g for one tablespoon of that food. A spoon of another ingredient cannot automatically inherit this mass. [F5]

Divide a measured finished yield

A hypothetical well-mixed recipe has an estimated retained total of 2,400 kcal and 120 g protein, with 1,600 g finished edible yield. A 300-g portion is 300 / 1,600 of the batch: **450 kcal and 22.5 g protein**. Eight genuinely equal portions would each weigh 200 g and provide an estimated 300 kcal and 15 g protein.

If only water is lost and the same retained energy remains while yield falls to 1,200 g, density rises from 1.5 to 2 kcal/g. A 300-g portion then contains **600 kcal**. If energy-containing material was discarded, the unchanged-total premise fails; reconsider the retained total before allocating it.

Added oil is not automatically all retained or all discarded

Adding 20 g oil contributes roughly **180 kcal** under the general 9 kcal/g factor before retention is considered. That is not the same as the olive-oil record's more specific 176.8 kcal estimate for 20 g. Account for the chosen source and any oil left in the pan or drained away; do not assign the entire amount to one serving without justification. [P3,F5]

The calculations above are analytical examples. No batch was prepared, no participant was studied and no exact personal energy need follows from the answers.

A reusable portion record

Use this record when reproducibility is useful. Leave unavailable information explicitly unknown. The fields distinguish observation, calculation and judgment rather than turning an estimate into a recommendation.

Record field	What to preserve
Identity and purpose	Record ID; date; author; estimate eaten amount, compare food, divide recipe or plan purchasing.
Food and evidence	Exact product or recipe; source URL and FDC/label ID; retrieval date and version; nutrient data type.
Food form	Raw/cooked; dry/rehydrated; drained/undrained; edible part; skin/bone/shell; preparation.
Nutrient basis	Per 100 g or per label serving with serving grams; kcal and nutrient units; missing fields.
Measured amount	Tared edible mass; scale resolution; or food-specific volume/count conversion and its source.
Recipe adjustments	Edible ingredient masses; additions; discarded material; yield/retention source; no double adjustment.
Finished allocation	Measured edible yield and stage; portion mass or equal-portion count; mixture uniformity assumption.
Calculation and result	Formula; unrounded intermediate value; sensibly rounded final estimate with units.
Uncertainty and scope	Estimated or unmeasured items; major variation; limits on use; no automatic adequacy or medical claim.
Successor history	Previous record ID; changed ingredients/process/source; reviewer and date; keep the prior version.

Keep these three outputs visibly separate

Measured quantity: what was weighed or observed.
Calculated estimate: the result derived from compatible source values.
Recommended intake: not generated by this template; requires a separate justified assessment.

Example status line: "Calculated from the named source and stated assumptions; not laboratory-tested; not a personal intake target." A proposed protocol or app rule requires separate manual ratification; implementation and deployment require their applicable authorizations.

Zen review one on nutrient claims

This review asks whether a true number has acquired a stronger meaning than its evidence supports. It is an internal desk analysis, not a nutrition intervention or validation study.

POINT

Claim under review: a daily energy total or a macronutrient split can describe part of a diet, but cannot by itself establish nutrient adequacy, an appropriate individual intake or a health outcome.

BRIDGE

FDA's 2,000-kcal reference supports label interpretation, not a universal personal requirement. The National Academies' RDA, AI and AMDR answer different questions. The current US dietary guideline's protein range is a policy target, while the adult protein RDA is a requirement-based reference. Keeping the source type and population attached prevents these numbers from silently replacing each other. [P1,N1,N2,N3]

PATH

Alternative A: retain calories only. It is simpler for energy accounting but cannot answer whether protein, essential fatty acids, fiber or micronutrients are adequate.
Alternative B: retain calories plus selected nutrients and pattern context. It answers more of the question but still does not diagnose deficiency or predict an individual outcome.
Stop condition: the record is insufficient for a personal diet or supplement decision when the relevant life stage, medical context or nutrient evidence is missing. Do not fill that gap with a universal target.

SHADOW

Counterexample: two foods can each provide 200 kcal but very different protein, fiber and micronutrients. A daily total can match 2,000 kcal while supplying little of an essential nutrient. Conversely, a nutritious pattern may supply a different energy total because needs vary. Therefore "same calories" does not imply "same nutrition".

A further counterexample is a person whose intake on one day is below an RDA. That observation alone does not establish clinical deficiency. An RDA is a reference for usual intake in a stated population, not a diagnostic test or a per-meal threshold. [N1]

Disposition

Preserve the calorie estimate and its units, but narrow any claim of adequacy. Keep reference type, age/life-stage scope and evidence limits visible. No algorithmic diet rule is authorized by this review.

Zen review two on portion integrity

This review asks whether the calculation still describes the food actually weighed. The arithmetic may be correct while the input identity is wrong.

POINT

Claim under review: nutrient values can be scaled by mass when the weighed edible food matches the source form and the calculation preserves the nutrient basis, units and recipe assumptions.

BRIDGE

USDA documentation supports per-100-g scaling and food-specific portion conversions. Yield and retention resources distinguish prepared edible food from as-purchased inputs and describe losses during preparation. They do not make every cup equal, every recipe homogeneous or every cooking process identical. [P4-P7]

PATH

Alternative A: use a matching cooked-food record and a cooked edible weight; it is direct when the source closely matches.
Alternative B: use ingredient contributions, justified retention adjustments and measured final yield; it better preserves a particular recipe but has more uncertain inputs.
Alternative C: use a documented rough household estimate when precision is unnecessary; communicate the lower confidence.
Stop condition: if food form, denominator or finished yield is unknown, do not present a precise result as established. Resolve the mismatch or keep a bounded estimate with its missing inputs.

SHADOW

Counterexample: 150 g of cooked rice is multiplied by a dry-rice energy value. The division is mathematically flawless but estimates a different food. Another: soup loses water and the old kcal-per-gram value is retained. A third: eight bowls weigh the same, but one contains most of the meat. All three invalidate the intended inference without requiring a formula error.

Label rounding creates another boundary. A zero-calorie label declaration may be permitted for a sufficiently small amount; multiplying the displayed zero by many servings does not prove chemical absence. A missing nutrient field is also not zero. [P3,P4]

Disposition

Require food identity, edible mass and source basis before calculation; measured final yield before recipe allocation; and a visible homogeneity/retention assumption. Preserve historical records instead of silently overwriting them after a recipe change.

Zen review three and adversarial audit

POINT

Claim under review: a useful food pattern can support nutrient adequacy without requiring every meal to be optimized or every bite to be weighed. A measurement method should remain a tool, not become a compulsory behavior.

BRIDGE

Dietary guidance emphasizes overall patterns and nutrient-rich food variety. Nutrient references vary across life stages and are not disease-treatment instructions. Individual preferences, culture, affordability, allergies and access affect feasible choices. A numerical calculation alone does not evaluate these constraints. [N1,N3,N4]

PATH

Use a qualitative pattern review for broad food variety; use labels or occasional measured portions for a specific uncertainty; use professional assessment when medical or life-stage needs require it. More intensive records can improve reproducibility but also add burden and false precision. Stop escalating tracking if it is unhelpful, distressing or being used to override hunger, medical advice or adequate access to food.

SHADOW

A technically varied menu may contain an allergen. A costly "ideal" menu may not be accessible. A high-protein pattern may still omit fiber or essential fats. A person can meet a spreadsheet target while becoming more distressed about eating. These counterexamples defeat the claim that a nutrient score alone certifies suitability.

Death Knell bounded counterexample audit

This audit tries to falsify the broadest claims using hypothetical cases only. It does not test people or authorize a revised live protocol.

The counterexamples reject five shortcuts: treating 2,000 kcal or a label serving as a prescription; treating essential fats as a mandatory oil; treating a policy protein goal as a new RDA or supplement dose; holding recipe energy constant despite nutrient-bearing losses; and ignoring historical source dates or missing fields. Each failure narrows the claim rather than authorizing a new rule.

Surviving narrow claim: matched food data and explicit assumptions support a reproducible estimate. **Not established:** a personal calorie requirement, clinical outcome, universal safe diet or complete nutrient adequacy. Scope is desk reasoning and arithmetic only. Reference ratification and any implementation or deployment authorization remain separate decisions.

Sources for nutrition foundations

Sources were checked on 23 October 2026 UTC. A retrieval date is not a publication date. Source numbers identify the evidence used; calculation examples and the Zen reviews are explicitly analytical synthesis.

[N1] NIH ODS Nutrient Recommendations and Databases

Definitions and limits of DRI and label reference values.

https://ods.od.nih.gov/HealthInformation/nutrientrecommendations.aspx

[N2] National Academies Macronutrient DRI Summary

Established reference tables pp 1-4; historical report, not a 2026 revision.

https://www.nationalacademies.org/cdn/materials/9fb9fae1-63a0-4048-88ad-3f972639149a

[N3] USDA and HHS Dietary Guidelines for Americans 2025-2030

Current policy edition. Printed pp 2-3 and 5: protein goal, produce, saturated fat and sodium.

https://cdn.realfood.gov/DGA.pdf

[N4] WHO Healthy Diet

Updated 26 January 2026. Fat replacement, limits and free sugars.

https://www.who.int/news-room/fact-sheets/detail/healthy-diet

[N5] National Academies Dietary Reference Intakes for Energy

2023 report summary; energy estimation and sources of variation.

https://www.nationalacademies.org/read/26818/chapter/2

[N6] NIH ODS Omega 3 Fatty Acids

ALA, EPA and DHA distinctions. The page's older total-fat DV is not used.

https://ods.od.nih.gov/factsheets/Omega3FattyAcids-HealthProfessional/

[N7] MedlinePlus Amino Acids

Reviewed 21 January 2025. Essential amino acids and protein variety.

https://medlineplus.gov/ency/article/002222.htm

[N8] FAO Dietary Protein Quality Evaluation in Human Nutrition

2013, pp 20-23. DIAAS framework, not a complete food-health ranking.

https://www.fao.org/4/i3124e/i3124e.pdf

[N9] WHO Carbohydrate Intake for Adults and Children

2023 guideline. Adult naturally occurring fiber recommendation.

https://www.who.int/publications/i/item/9789240073593

Sources for life stage and portion method

[N10] NIH ODS Vitamin B12

Table 1 and food-source discussion.

https://ods.od.nih.gov/factsheets/VitaminB12-HealthProfessional/

[N11] NIH ODS Iron

Table 1 and bioavailability discussion.

https://ods.od.nih.gov/factsheets/Iron-HealthProfessional/

[N12] NIH ODS Calcium

Table 1 and food-source discussion.

https://ods.od.nih.gov/factsheets/Calcium-HealthProfessional/

[N13] NIH ODS Vitamin D

Table 2; minimal-sun assumption and food sources.

https://ods.od.nih.gov/factsheets/VitaminD-HealthProfessional/

[N14] NIH ODS Folate

Table 1; dietary folate equivalents and life-stage distinctions.

https://ods.od.nih.gov/factsheets/Folate-HealthProfessional/

[N15] NIH ODS Iodine Consumer Fact Sheet

Iodized salt and variable food sources.

https://ods.od.nih.gov/factsheets/Iodine-Consumer/

[N16] National Academies Water Reference Report

2005 report chapter 4. Total water includes food and beverages.

https://www.nationalacademies.org/read/10925/chapter/6

[N17] FDA Major Food Allergens

Current US labeling list, not all possible allergies.

https://www.fda.gov/industry/fda-basics-industry/what-major-food-allergen

[N18] FDA Dietary Supplements

No FDA premarket approval of supplement safety and effectiveness.

https://www.fda.gov/consumers/consumer-updates/fda-101-dietary-supplements

[P1] FDA What Is on the Nutrition Facts Label

Serving size, label basis and the 2,000-Calorie reference.

https://www.fda.gov/food/nutrition-facts-label/whats-nutrition-facts-label

[P3] 21 CFR 101.9 Nutrition Labeling of Food

Current eCFR through 21 October 2026, sections c1-c2; factors and rounding.

https://www.ecfr.gov/current/title-21/chapter-I/subchapter-B/part-101/subpart-A/section-101.9

Sources for measurement and food records

[P4] USDA Foundation Foods Documentation

Per-100-g scaling, portion weights and missing data; not the six examples' data type.

https://fdc.nal.usda.gov/Foundation_Foods_Documentation/

[P5] USDA Food Buying Guide About

As-purchased and edible/prepared yield. Program crediting is not an adult prescription.

https://foodbuyingguide.fns.usda.gov/Home/About

[P6] USDA Nutrient Retention Factors

Release 6, 2007. Historical process-specific factors, not new recipe measurements.

https://www.ars.usda.gov/northeast-area/beltsville-md-bhnrc/beltsville-human-nutrition-research-center/methods-and-application-of-food-composition-laboratory/mafcl-site-pages/nutrient-retention-factors/

[P7] USDA Research Description on Nutrient Retention

2008 research description; true-retention and yield distinction.

https://www.ars.usda.gov/research/publications/publication/?seqNo115=225934

[SLEG] USDA Documentation on SR Legacy Provenance

Preface identifies the final April 2018 SR Legacy release and its non-updating status.

https://fdc.nal.usda.gov/Foundation_Foods_Documentation/

[F1] USDA FDC 168877 Rice raw enriched

SR Legacy; portal publication 1 April 2019; exact description on page 10.

https://fdc.nal.usda.gov/food-details/168877/nutrients

[F2] USDA FDC 168878 Rice cooked enriched

SR Legacy; portal publication 1 April 2019; exact description on page 10.

https://fdc.nal.usda.gov/food-details/168878/nutrients

[F3] USDA FDC 173735 Black beans boiled without salt

SR Legacy; portal publication 1 April 2019; exact description on page 10.

https://fdc.nal.usda.gov/food-details/173735/nutrients

[F4] USDA FDC 173424 Egg whole hard boiled

SR Legacy; portal publication 1 April 2019; exact description on page 10.

https://fdc.nal.usda.gov/food-details/173424/nutrients

[F5] USDA FDC 171413 Olive oil salad or cooking

SR Legacy; portal publication 1 April 2019; exact description on page 10.

https://fdc.nal.usda.gov/food-details/171413/nutrients

[F6] USDA FDC 171688 Apples raw with skin

SR Legacy; portal publication 1 April 2019; exact description on page 10.

https://fdc.nal.usda.gov/food-details/171688/nutrients